

## Unit 05: Scales and Maps

Name: \_\_\_\_\_ ( )

Class: S2-0 \_\_\_\_

### Big Ideas: Ratio and proportion, Similarity

#### Unit Essential Understanding

1. Geometry reflects natural phenomena that could be modeled mathematically.
2. A ratio is the relationship of two or more quantities or measurements.

#### Unit Essential Questions

1. How are ratio and scale related mathematically?
2. How do you determine the real distance of 2 towns from the Google map?

References: • Think! Mathematics New Syllabus Mathematics 2B, 8th Edition Chapter 8, section 8.3 on Similarity and Enlargement.

### Teaching To The Big Idea ...

| Lesson sequence in the unit                |                                                |            |           |          |          |             |                 |        |
|--------------------------------------------|------------------------------------------------|------------|-----------|----------|----------|-------------|-----------------|--------|
| Student Learning Outcomes                  | Dimensions (Please tick the appropriate boxes) |            |           |          |          |             |                 |        |
|                                            | FUNCTIONS                                      | INVARIANCE | NOTATIONS | DIAGRAMS | MEASURES | EQUIVALENCE | PROPORTIONALITY | MODELS |
|                                            | F                                              | I          | N         | D        | M        | E           | P               | M      |
| Similarity, Enlargement and Scale Drawings |                                                | ✓          | ✓         | ✓        | ✓        |             | ✓               | ✓      |

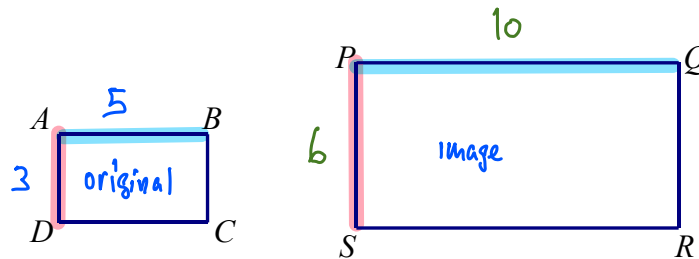
### Checklist for Self-Assessment (Please tick ✓ at the appropriate boxes)

| Levels  | Characteristics                                                                             | Knowledge and Skills                                              | I need help.<br>I need to see an example. | I can do this on my own. | I can do this on my own as well as explain my solution to my classmates or teacher. |
|---------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------|--------------------------|-------------------------------------------------------------------------------------|
| Level 0 | Reproduction of facts, rules, formulae;<br>No explanations required                         | Identify map scales involving direct proportion                   |                                           |                          |                                                                                     |
| Level 1 | Procedural tasks without connections,<br>Algorithmic in nature;<br>No explanations required | Solve equations involving proportion of lengths and areas on maps |                                           |                          |                                                                                     |

## Similarity and Scale Factor

Same shape, different size.

Two figures are **similar** if they have the **same shape** and their **dimensions are of a common ratio**, known as the scale factor.



The figure above shows rectangles  $ABCD$  and  $PQRS$  with  $AB = 5$ ,  $AD = 3$ ,  $PQ = 10$  and  $PS = 6$ .

Note that the ratio  $\frac{PQ}{AB} = \frac{PS}{AD} = 2$ . We then say that rectangle  $PQRS$  is an enlargement of rectangle  $ABCD$  with scale factor 2.

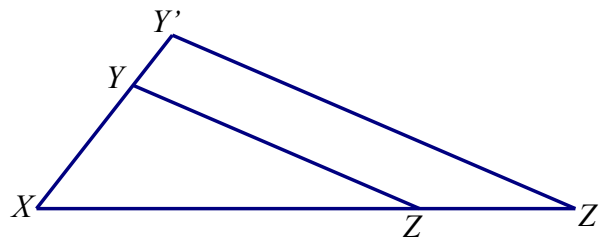
A scale factor shows how much a shape is enlarged or reduced.

The formula to calculate a scale factor is  $\frac{\text{Dimension of new shape/image}}{\text{Dimension of original shape}}$ .

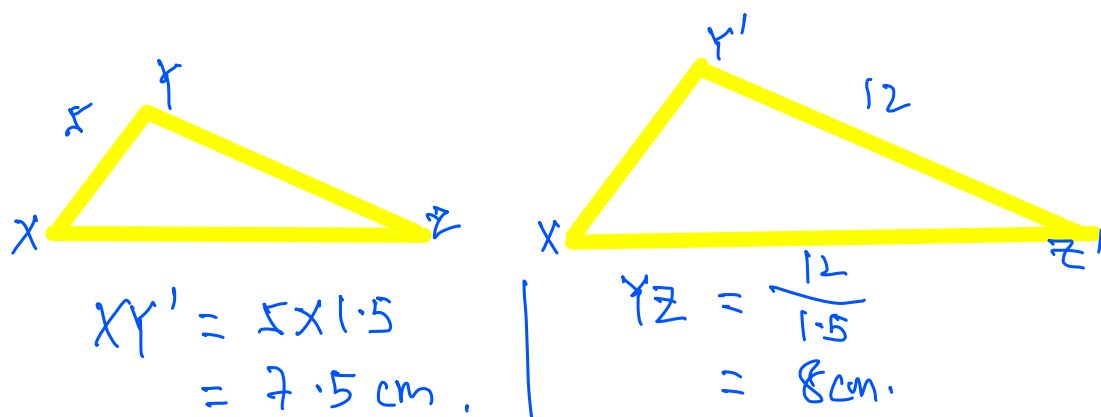
If the scale factor is greater than 1, the figure enlarges; if it is between 0 and 1, the figure reduces in size.

### Example 1

In the figure,  $\triangle XY'Z'$  is an enlargement of  $\triangle XYZ$  with a scale factor of 1.5.



Given that  $XY = 5$  cm and  $Y'Z' = 12$  cm, find the length of  $XY'$  and of  $YZ$ .



In our daily activities, we sometimes need to enlarge or reduce pictures of actual objects. For example, a contractor or interior designer would need a floor plan of the actual apartment unit before carrying out the home renovation work.

### Example 2

The scale of the floor plan of a house is 1 cm to 2.5 m.

Find

- (i) the actual length of the dining room if it is represented by a length of 1.25 cm on the floor plan.
- (ii) the length on the floor plan that represents the width of a bedroom if its actual width is 3.4 m.

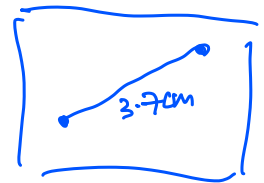
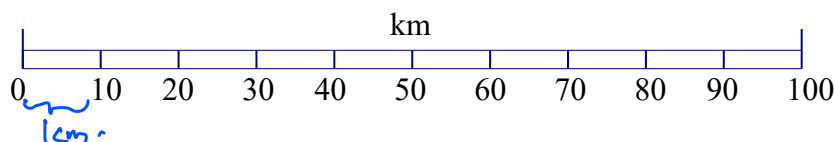
$$\begin{aligned} \text{(i) Actual} &= 1.25 \times 2.5 \\ &= \underline{3.125} \text{ m} \end{aligned}$$

$$\begin{aligned} \text{(ii) Length on floor plan} &= \frac{3.4}{2.5} \\ &= \underline{1.36} \text{ cm} \end{aligned}$$

### Map Scales

Maps are scaled down drawings of actual lands. The scale of a map is usually given at a corner of the map, in the form of ratio, fraction or scale bar.

Here is an example of map scale bar:



If you use a ruler to measure the length from 0 to 10 km and the reading is 1 cm, then we say 1 cm represents 10 km.

As 10 km = 10 × 1000 × 100 = 1000000 cm, the scale of the map is 1: 1000000.

$$\begin{aligned} 1 \text{ km} &= 1000 \text{ m} \\ 1 \text{ m} &= 100 \text{ cm} \end{aligned}$$

The scale of 1:  $n$  can also be represented as a **representative fraction (R.F.)** of  $\frac{1}{n}$ .

2 cm represents 5 km  
2 cm represents 500000 cm, 1 cm represents 250000 cm.

If a map has a scale of 2 cm to 5 km, then its R.F. is  $\frac{1}{250000}$  (which is  $\frac{2}{5 \times 1000 \times 100}$ ) or it has a scale of 1: 250000 (where both sides are of same units).

**Remark:** The **numerator of R.F. must always be 1**.

and  
↓  
conditions  
for R.F.  
Write as fraction  
 $\frac{1}{250000}$

### Example 3

Given that the scale of a map is 1 cm: 25 km, express it in the following forms:

- Same units → (a) 1: r  
→ (b)  $\frac{1}{r}$

1 cm represents 25 km

RF

(a) 1 cm represents 25 km

1 cm represents  $25 \times 1000 \times 100$

1 cm represents 2500000

1 : 2500000

(b)  $\frac{1}{250000}$

### Example 4

Given that the scale of a map is  $\frac{1}{15000}$ , find the distance represented on the map for the following:

- (a) 750 m  
(b) 3.9 km

R.F.

1 cm represents 15000 cm?

← same

1 m represents 15000 m?

→ 100 cm represents 1500000 cm  
1 cm represents 15000 cm

(a) 1 cm represents 15000 cm

1 cm represents 150 m

750 m is represented by  $\frac{750}{150} = 5$  cm

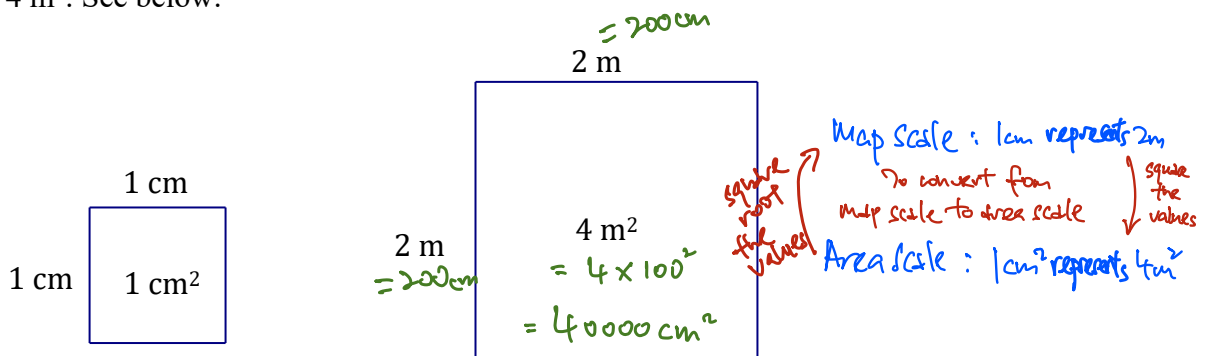
It does not matter  
but we choose  
based on the  
units in the question.

(b) 1 cm represents 150 m  
3900 m is represented by  $\frac{3900}{150} = 26$  cm

### Area Scales

We can find the actual area of a site from its area on the map.

For example, if the scale of a map is 1 cm to 2 m, then 1 cm<sup>2</sup> on the map represents an actual area of (2 m)<sup>2</sup> = 4 m<sup>2</sup>. See below.



Therefore, the area scale of the map is 1 cm<sup>2</sup> to 4 m<sup>2</sup> when the map scale is 1 cm to 2 m.

That is the area scale is 1: 40000 when the map scale is 1:200.

Length  
conversion of units  
1 m = 100 cm

1 km = 1000 m

Area  
conversion of units  
1 m<sup>2</sup> = 100<sup>2</sup> cm<sup>2</sup>

1 km<sup>2</sup> = 1000<sup>2</sup> m<sup>2</sup>

If a map scale is  $1:r$ , then its area scale is  $1:r^2$ .

square the values

square root the values

### Example 5

Given that the scale of a map is  $1:35\,000$ , find

- (a) the **actual** area, in  $\text{km}^2$ , of an area which is  $12\text{ cm}^2$  on the **map**,  
 (b) the map area in  $\text{cm}^2$ , of an area which is  $28\text{ km}^2$ .

(a)  $1\text{ cm}$  represents  $35000\text{ cm}$   
 Map scale:  $1\text{ cm}$  represents  $0.35\text{ km}$   
 Area scale:  $1\text{ cm}^2$  represents  $0.35^2\text{ km}^2$   
 $1\text{ cm}^2$  represents  $0.1225\text{ km}^2$   
 $\therefore 12\text{ cm}^2$  represents  $12 \times 0.1225$   
 $= 1.47\text{ km}^2$

(b)  $1\text{ cm}^2$  represents  $0.1225\text{ km}^2$   
 $28\text{ km}^2$  is represented by  
 $\frac{28}{0.1225} = 228\frac{4}{7}\text{ cm}^2$   
 exact value is preferred than 3 s.f.

### Example 6

Given that  $2\text{ cm}$  represented  $5\text{ km}$  on a certain map, find the

- (a) scale of the map in the form  $1:r$ ;  
 (b) scale of the map in the form  $\frac{1}{r}$ ;  
 (c) actual distance between two police stations that are  $6.6\text{ cm}$  apart on the map;  
 (d) distance between two libraries as represented on the map if they are actually  $9\text{ km}$  apart on the ground.

(a)  $2\text{ cm}$  represents  $5\text{ km}$   
 $2\text{ cm}$  represents  $500000\text{ cm}$   
 $1\text{ cm}$  represents  $250000\text{ cm}$   
 $\therefore 1:250000$   
 (b)  $\frac{1}{250000}$

(c)  $1\text{ cm}$  represents  $250000\text{ cm}$   
 $6.6\text{ cm}$  represents  $6.6 \times 250000\text{ cm}$   
 $= 16.5\text{ km}$   
 (d)  $2\text{ cm}$  represents  $5\text{ km}$   
 $\frac{2}{5} = \frac{\text{map distance}}{9}$   
 Ans:  $\frac{9}{5} \times 2 = 3.6\text{ cm}$

The scale of a map is 1: 40000.

- (a) If a road is represented in 2.5 cm on the map, find the actual length of the road, in km.  
(b) If the actual area of a plot of land is  $0.5 \text{ km}^2$ , find its area on the map, in  $\text{cm}^2$ .

(a) 1 cm represents 40000 cm

1 cm represents 0.4 km

$$2.5 \text{ cm to } 0.4 \times 2.5 \text{ km} \\ = 1 \text{ km.}$$

Square the values

(b) Area scale :  $1 \text{ cm}^2$  represents  $0.66 \text{ km}^2$   
 ————— represents  $0.5 \text{ km}^2$

$$\therefore \frac{0.5}{0.16} = \underline{0.3125 \text{ cm}^2}$$

### Example 8

The actual area of a pond which is  $16 \text{ km}^2$  is represented on map A by an area of  $400 \text{ cm}^2$ .

- (a) If map  $A$  is drawn to a scale of  $1:k$ , find the value of  $k$ .
- (b) The distance between two locations on Map  $A$  is 8.6 cm. Find the distance of the same two locations on map  $B$  which has a scale of 1 cm: 0.4 km.
- (c) State which map has a longer map distance of the two locations.

(a) Area scale :  $400 \text{ cm}^2$  represents  $16 \text{ km}^2$

Map scale : 20 cm represents 4 km ↘

20 cm represents 40000 cm

1 cm represents 20000 cm

$$\therefore K = 20000$$

convert from  
tree scale to map scale  
(square root values)

(b) Map A: 8.6cm represents  $8.6 \times 20000$  cm

$$= \frac{1.72 \text{ km.}}{\text{actual.}}$$

Map B: 1 cm represents 0.4 km  
— represents 1.72 km

$$\frac{1.72}{0.4} = 4.3 \text{ cm.}$$

(c) MspA

1 cm represents 0.2 km  $\rightarrow$  2 cm represents 0.4 km

Map B

1 cm represents 0.4 km

Ans: Map A.

On Map A, 2cm represents 0.4km  
while on Map B, 1cm represent 0.4km

## **Summary**

### **Similarity and Scale factor**

Two figures are similar if they have the same shape and their dimensions are of a common ratio, known as the scale factor.

The formula to calculate a scale factor is  $\frac{\text{Dimension of new shape}}{\text{Dimension of original shape}}$ .

### **Map Scale**

A map scale defines the ratio between distance on a map and actual distance on the ground.

Map scale can be represented in:

1. ratio form  $1:r$ , using same units on both sides or
2. representative fraction (R.F.), of the form  $\frac{1}{r}$ .

### **Areas on a Map**

If the map scale is  $1:r$ , then its area scale is  $1:r^2$ .